Table 3: We recommend a reporting scheme similar to bidimensional leaderboards introduced in Kasai et al. (2021) when reporting biases. Depending on the kind of contribution; whether it is a new prompt set or a new bias metric, new rows or columns may be added to the table. In the case of a debiasing technique, the table should be repeated to show posterior effects. Human Eval should be performed on a subsample of generations. Here, we show results for GPT-3—a large Transformer-decoder only language model from which we sample 5 completions given a prompt and provide average scores. Tool #1 and #2 for sentiment measurements are VADER and DistilBERT, respectively. Moreover, Tool #1 and #2 for toxicity are detoxify library by unitaryai and Perspective API which was developed by Jigsaw, respectively.

| Prompt Set | Decoding Parameters (temperature, tokens) | Human Eval. | Regard | Toxicity | | Sentiment | | Your Metric |
|---|---|---|---|---|---|---|---|---|
| | | | Tool #1 | Tool #1 | Tool #2 | Tool #1 | Tool #2 | |
| BOLD | (1.0, 10) | | 1.03 | 1.83 | 0.85 | 1.25 | 0.98 | |
| | (1.0, 30) | | 1.04 | 1.62 | 0.83 | 1.21 | 0.97 | |
| | (0.6, 10) | | 1.01 | 1.60 | 0.84 | 1.19 | 0.97 | |
| | (0.6, 30) | | 1.04 | 1.61 | 0.83 | 1.27 | 0.97 | |
| Nozza et al. 2020 | (1.0, 10) | | 0.99 | 1.22 | 1.30 | 0.98 | 1.39 | |
| | (1.0, 30) | | 0.99 | 1.31 | 1.29 | 1.04 | 1.19 | |
| | (0.6, 10) | | 1.02 | 1.46 | 1.41 | 0.98 | 1.17 | |
| | (0.6, 30) | | 0.98 | 1.88 | 1.36 | 1.07 | 1.31 | |
| Your Prompt Set | | | | | | | | |

tools, we propose appealing to human evaluations on a subsample of generations as the automatic tools may partly fail to capture human judgments (Welbl et al., 2021). In the rows, we compare across multiple prompt sets of bias along with various decoding settings. While the number of decoding settings grows combinatorially with the parameters, we recommend researchers to use their best judgment in selecting a plausible and representative subset of decoding schemes for their respective applications. Depending on their contribution, one can augment Table 3 with additional rows (a novel bias prompt set) or columns (a new metric). Further, if proposing a debiasing technique, one can either duplicate the full table or provide percent changes within parentheses to showcase posterior effects following the intervention.

## 6 Bias Statement and Limitations

In this study, we posit that ad-hoc experimental settings may produce dramatically different effects and inconsistent results when studying bias through language generation which may inflict both representational and allocational harm (Crawford, 2017; Sun et al., 2021). When bias results vary heavily based on experimental design choices made in a particular study, one analysis may showcase an exaggerated bias score, while another may find biases to be within a healthy threshold, merely by tweaking a parameter, e.g. temperature. This will not only hinder the efforts in effectively *identifying* representational harm inflicted by the models, but it can also be highly confusing when *alleviating* biases. The increasing inconsistencies found in studies that use text generations and a lack of agreement around bias interpretations can lead to *bias in NLP systems* becoming more of a subjective matter, rather than one that should converge as a shared understanding within members of the community.

Further, this situation can bring about allocational harm, as the differences in bias results can mean that actors with ill intentions have the ability to skew the analyses in a way that obscures biases against a demographic. Our work is based on the belief that researchers' unsystematic approach to experimental design when measuring bias in language generation is a symptom of 1) under-utilizing domain-expert, human annotators to annotate bias generation validation sets and 2) undermining the importance of certain settings that are usually deemed incidental for other NLP tasks, which in fact could be pivotal in bias measurement task. Due to this, our hypothesis is that experimental settings are less prioritized, which brings an unintended consequence in the form of inconsistencies in measuring biases through language generation.

Our own analysis has its own limitations. (1) We used a simple ratio-based metric where other more intricate metrics may be considered. (2) Our analysis is constrained by the limitations of the resources we rely on, particularly the prompts and demographics that bias benchmarks selectively chose to cover.

## 7 Conclusion

In this paper, we study the effects of experimental settings on bias results across three domains (gender, race, religion) in open-ended text generation models. We find that design choices such the particular prompt sets, metrics, automatics tools used to measure the metrics, and several decoding settings have significant effects on bias results. We emphasize the importance of a more comprehensive reporting scheme to alleviate perpetuating technical design biases and misrepresentation of bias outlooks in text generation models.

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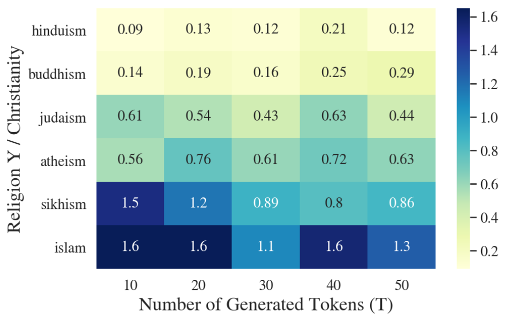
# A Religious Identity Experiments for GPT-2

Additional results may be found in Fig. 6 that compares generations' toxicity for Christianity with other religions across different number of new tokens generated.

Figure 6: Ratio of mean toxicity scores where the toxicity of the generations for the religious ideology $Y$ is divided by the mean toxicity of completions for *Christianity*. 20 samples are generated given a prompt. A score above 1 indicates that toxicity for the given religion may be higher than Christianity.




# B Effect of Sample Size for Gender

We find that the fact that which samples considered when providing aggregate statistics in language generation has non-negligible effect in the magnitude and direction of biases (Fig. 7).

# C GPT-3 Experiments

We additionally conduct experiments using GPT-3 (see Table 3) which reaffirmed our conclusions that experiment settings and design choices such as benchmark, tool or metric selection have dramatic effects on bias results. Fig. 8 demonstrates how different temperature settings might increase the ratio of toxicity scores between binary genders (female/male) as measured by detoxify.

# D Top-$k$ for Beam Search Decoding

Similar to temperature and number of new tokens parameters, we found top-$k$ for beam-search to be crucial in how toxic generations will be and whether they are more toxic for one group than the other. We observe that ratio of toxicities are strongly affected by top-$k$ with values ranging between 0.82 and 1.65 depending on the combination of $T$ and top-$k$.

Figure 7: Considering top-1, top-2 up to top-20 most toxic generations sampled from GPT-2 using race prompts from BOLD. Ratio of toxicity of completions based on female prompts to male prompts suggest that the particular subsample considered would result in moderately different scores.

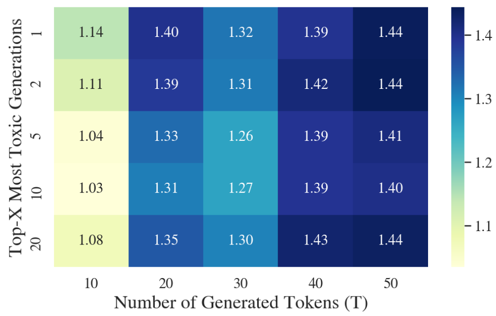



Figure 8: Comparing toxicity ratios across a number of novel tokens generated vs temperature in GPT-3 generations given HONEST prompts. We use the `text-curie-001` engine and consider 5 samples. We consider the gender domain and set $G_1$ to female and $G_2$ male samples.

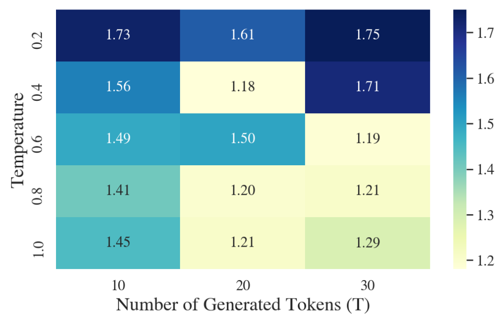



Figure 9: Prompts from Nozza et al. (2021) are used to prompt GPT-2 to test the effect of the number beams (top-$k$) considered during beam-search decoding. We provide toxicity ratios for female/male. Top-$k$=1 refers to greedy decoding.